

Survey of next-generation optical wireless communication technologies for 6G and Beyond 6G

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ABSTRACT

Demand for high-speed services has surged. The increasing congestion in the radio frequency system, which ranges from 3 kHz to 300 GHz, threatens the growth of next-generation wireless technologies. Despite advancements in signal processing and dedicated modulation techniques, accommodating users' growing demands with current frequency bands will be a formidable challenge in the coming decades. However, an unlicensed spectrum could potentially offer data rates from 10 Gbps to Tbps for communication systems in the future. Optical Wireless Communication (OWC) technologies can address the limitations in communication bandwidth associated with traditional radio frequency systems. This survey paper provides an overview of the technology and reviews optical wireless communication technologies related to 6 G and beyond 6 G systems. We examine the key technologies essential for OWC and outline the latest developments in areas such as classification, spectrum usage, system architectures, and applications. IEEE 802.15.7 standardization is also discussed in this survey paper, along with some experimental results to highlight the impact of Deep Learning on OWC system. Additionally, we discuss future research challenges for OWC systems.

1. Introduction

The sixth generation (6 G) of mobile communication represents the latest telecommunication generation, providing advanced features such as high speed communication, high connectivity, very low latency, high security, and very good quality of services [1]–[6]. Consequently, future networks must be equipped to support high communication capacity, thus reducing power consumption and latency [1–6]. To enable these ultra-dense, ultra-fast admission networks, robust ultra-capacity connectivity is crucial for 6 G and beyond [6].

With the increasing adoption of smart home/factory systems, the smart devices is being connected to the Internet required to have high speed [7–11]. At present, the radio frequency (RF) spectrum has many different applications [12]. However, the RF spectrum, particularly below 40 GHz, is inadequate for meeting the increasing capacity demands for 6 G and 6 G beyond technologies and the Internet of Thing (IoT) paradigm. The present RF spectrum is nearly exhausted due to its favorable communication properties, such as low attenuation and wide coverage. This, coupled with regulatory constraints and significant interference among adjacent RF access points, means that the massive

connectivity requirements of future mobile data traffic cannot be met by current wireless technologies.

Therefore, researchers are exploring alternative spectra, including millimeter and nanometer waves, for wireless communication [13]. RF communication technology has seen significant growth in recent years. Qualcomm predicts that future wireless systems will require a thousand-fold increase in wireless communication capacity. As a result, the RF spectrum demand will continue to increase. To alleviate this pressure, an innovative solution that operates in unlicensed spectrum regions, such as optical wavelengths, is required. The light waveform has higher spectrum compared with electromagnetic waveform. It is free from electromagnetic interference (EMI) and have higher bandwidth compared RF waveform. Additionally, the light waveform offers outstanding line-of-sight separation and enables cost-efficient augmented reality experiences. Accordingly, optical wireless communication (OWC) is a crucial emerging approach for next-generation wireless connectivity. In this context, the light spectrum is being considered as a substitute for the RF band, with applications such as visible light communication (VLC), Light Fidelity (LiFi), and optical camera communication (OCC). The advantages of light waves over the

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radio spectrum are outlined below:

- The visible light spectrum is safe for humans [14] and does not have the adverse effects associated with radio waves.
- Light spectrums offer significantly greater bandwidth than the RF spectrum, exceeding it by more than 1000 times.
- Light waves are safer and more efficient than RF waves.

The IEEE Standards Association has published VLC technology standards that include some physical layer (PHY) specification modes unique to the VLC system [15]. Additionally, the Short Range Optical Wireless Communications Standard [16] has been introduced for modifying IEEE 802.15.7–2011, incorporating some PHY specification modes dedicated to OCC technology. This updated standard features some OCC modulation schemes designed for short-range OCC systems, such as the high frame-rate approach, region of interest (RoI) signaling, rolling shutter method, and screen or projector-based methods. Despite these advancements, the commercialization of OCC methods has been hindered by the absence of a suitable application for their use. Unlike LiFi, which uses photodiodes (PDs) to collect data from Light Emitting Diodes (LEDs), OCC employs cameras as receivers. Recent developments in image processing methods, including computer vision and artificial intelligence techniques, have made OCC technologies more advantageous than LiFi/VLC. These benefits include longer distances, easy integration for multiple input multiple output (MIMO) methods, and enhanced mobile effects of outdoor environments, such as drone-to-drone communication and vehicle-to-vehicle networks. Numerous corporations and academic institutions, after recognizing the feasibility of OWC technology, are aggressively pursuing its commercialization. For instance, Panasonic has established LinkRay, which uses OCC technology to transmit signals by analyzing IDs from light sources. Similarly, Picalio was introduced by Casio, the indoor positioning technology utilizing Casio's unique camera system, where LEDs transmit OCC signals through color signals. Both Panasonic and Casio have contributed to these advancements, as mentioned in [17]. To further enhance the data speed and communication range for OCC technology, researchers from Korea, Japan, and the United States have formed IEEE 802.15.7a Task Group (TG7a). This group aims to improve current OCC systems by incorporating advanced technologies, including orthogonal frequency-division multiplexing (OFDM), MIMO, non-orthogonal multiple access (NOMA), and hybrid modulation, potentially increasing the data speed and communication range of OCC system systems [17]. Besides some advantages of OWC, we also face with some problems to design OWC systems compared RF systems. OWC is revealed to be inefficient for NLoS (non-light-of-sight) transmission then the cover areas may be limited. From that, many researchers proposed some ideas to resolve these limitations. Currently, OCC among OWC standards has some limitations such as: the lack of bidirectional communication, which is crucial for enhanced reliability of OCC system. The next generation of OWC for 6 G and beyond should aim to overcome these limitations at the PHY and MAC layers by leveraging Artificial Intelligence (AI) to improve overall performance. We plan to develop a new OWC standard that addresses the shortcomings of previous versions, with a focus on supporting high mobility, improving performance in NLoS environments, incorporating AI-based PHY/MAC layers, and enabling joint communication and sensing.

The rest of the survey paper is structured as follows: Section II offers a comprehensive overview of OWC systems, including their arrangement based on receiver technologies, data rate, and communication range. Section III covers the IEEE 802.15.7 standards in detail. Section IV surveys the review and explores future research approaches for OWC technologies. Finally, the conclusions of the survey are presented in Section V.

2. System overview

2.1. Methodology

In this survey paper, numerous research papers related to OWC technologies were collected from IEEE Xplore, ScienceDirect, MDPI, and Google Scholar, etc. using keywords such as “OWC”, “LiFi”, “VLC”, “OCC”, “FSO”, and “IEEE 802.15.7 standard”. The selected papers focused on research articles published between 2001 and 2024, offering technical insights into OWC systems, applications, advantages, and limitations. These articles were reviewed, categorized, summarized, and analyzed to identify key trends, strengths, limitations, and future directions in OWC technologies, based on both technology type and application area. In addition, IEEE standards published between 2011 and 2025 were also categorized and summarized to explore potential directions for next-generation OWC standards.

2.2. Summary of OWC technologies

OWC refers to the use of visible light (VL), infrared (IR), or ultraviolet (UV) wave bands as the transmission medium for a communication system [17–23]. OWC systems that use the visible light waveform are generally referred to as VLC, which are capable of providing information transmission, illumination, and localization services simultaneously [24]. Moreover, Free-Space Optics (FSO) systems can use the IR/VL/UV spectrum [17,25]. The UV spectrum applications in particular support high-data-rate optical links for both NLOS and LOS scenarios, as proposed in [26–28]. This paper reviews the advantages in various OWC systems, such as VLC, LiFi, OCC, and FSO, to meet the 6 G and beyond 6 G requirements. While some of these technologies are already commercialized, others are still under development. High-speed data transmission in VLC is achieved using either LED light sources or laser diodes (LDs) [29–31]. LiFi [32–34], a complementary approach to WiFi, offers high data rate communication by using light spectrum (IR/V-L/UV). It typically uses LEDs/LDs at the transmitter side to transfer optical signals and photodetectors at the receiver side to collect the optical signals [34–36]. While VLC/LiFi systems are well-suited for high communication capacity for indoor environments, they are fewer suitable for outdoor environments due to non-support long-range communication. In [37–39], the authors proposed OCC, a branch of OWC that utilizes LEDs to transmit data and image sensors (IS) to receive data. OCC has the advantages of noninterference and high Signal-to-Noise Ratio (SNR) values in outdoor environments [38], and it performs consistently over extended communication distances [39]. FSO communication [40–44], which primarily utilizes the near-infrared (NIR) band, also supports UV and VL bands. FSO systems establish high-data-rate communication links using narrow beams of intense light between two points, with applications ranging from inter-chip to inter-satellite communication [40–44]. Moreover, [45] proposed a high data rate camera on-off keying (OOK) system, considering the rolling shutter method in a rolling shutter camera, but it has limitations, such as a short communication distance and poor performance in OCC systems. A high-speed OCC integrating OFDM technology and a rolling shutter camera was proposed in [46], referred to as RS-OFDM. Additionally, [47] introduced a MIMO technique for OCC systems to increase data rates, but it has the disadvantage of a large transmitter and high setup cost. By using OWC technologies instead of RF technologies, the communication system does not face with the interference effect. As we mentioned, the RF system transmit data by broadcasting radio signal, then the interference effect among RF systems make a big problem before designing the communication system. OWC technologies also are flexible deployment and low cost compared RF by using existing devices in the road or building (LED/LD light sources, camera, CCTV, etc.).

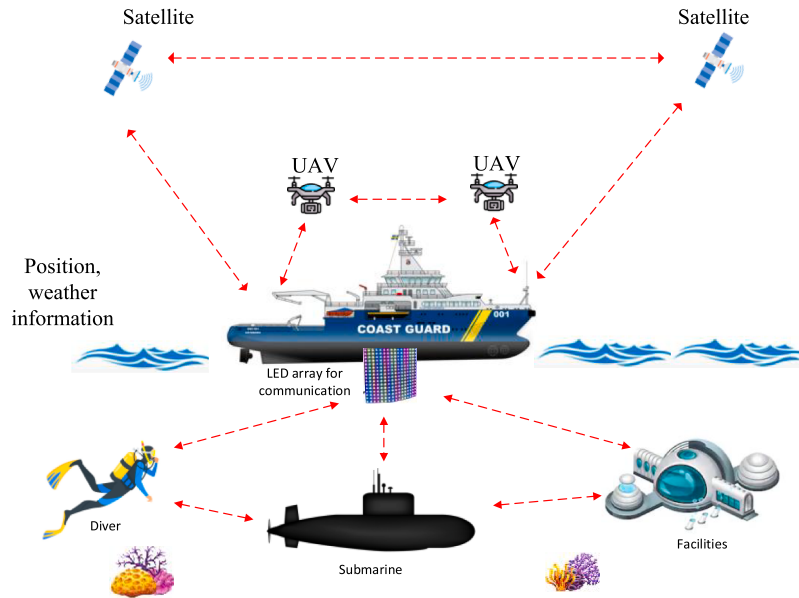


Fig. 1. OWC application for 6 G and beyond 6 G.

2.3. OWC applications

Fig. 1 illustrates the potential application of OWC technologies in drone communication, underwater communication, and satellite communication for 6 G and beyond 6 G systems. In these scenarios, Drone communication system, satellite communication system, and underwater communication system utilize information from smart lighting in coast guard, drones, satellite, underwater-facilities and submarine, which are collected optical signals by PDs and IS. Drone networks are an emerging research area with potential uses in army and civilian applications [48]. In these configurations, coast guard lights, submarine lights, drone strobe lights, and satellite light sources can be set up as OWC transmitters, thus lowering system costs. To communication satellite to satellite, FSO technologies should apply to take advantages of long distance and high speed communications of FSO. From satellite, we can connect with 6G/beyond 6 G base stations by RF technologies. While RF technology is recommended for communication between base stations (BS) and satellite, OWC is preferred for V2V, V2I, underwater communication services, and drone services due to its advantages over RF. During TG7a meetings, vehicular applications employing OCC methods with the rolling shutter method were recommended for the movement environment at a range of up to 30 m. The concept of laser uses for OCC, involving LDs instead of traditional LEDs, is projected to extend communication distances to up to ~several hundred km. This advancement is envisioned in next-generation OWC

technologies. As shown in Fig. 1, the OWC technologies can apply for 6G/ beyond 6 G with many applications and coverage areas. With long communication up to hundred km distance, the FSO technology reveal as a good candidate to transmit data such as Satellite to Satellite, Satellite to coast, etc. With some harsh channel conditions, OCC technologies can resolve the challenging channel (raining, snowing, high mobility, etc.). With a short distance and the high data requirement, VLC/LiFi should use to take in advance the high bandwidth of light spectrum instead of RF system.

2.4. Optical wireless communication for 6 G and beyond system

OWC offers promise for enabling high-speed 6 G systems and supporting the extensive connectivity needs of the IoT system. Next-generation 6 G and 6 G beyond technologies will require advanced capabilities to integrate ultra-dense heterogeneous networks. One simple but highly effective method to boost network capacity involves reducing cell sizes [49]. Current communication networks are increasingly incorporating linked small cells, such as picocells and femtocells, to achieve the requirements of 6 G and beyond 6 G systems. OWC technologies can propose high-dense services to address 6G/ beyond 6 G requirements. Additionally, OWC can serve as high communication capacity solutions to 6 G and beyond 6 G systems. OWC systems are characterized by low power consumption, which is a crucial feature for 6 G [50]. They also provide reliable connections, a top priority for 6 G

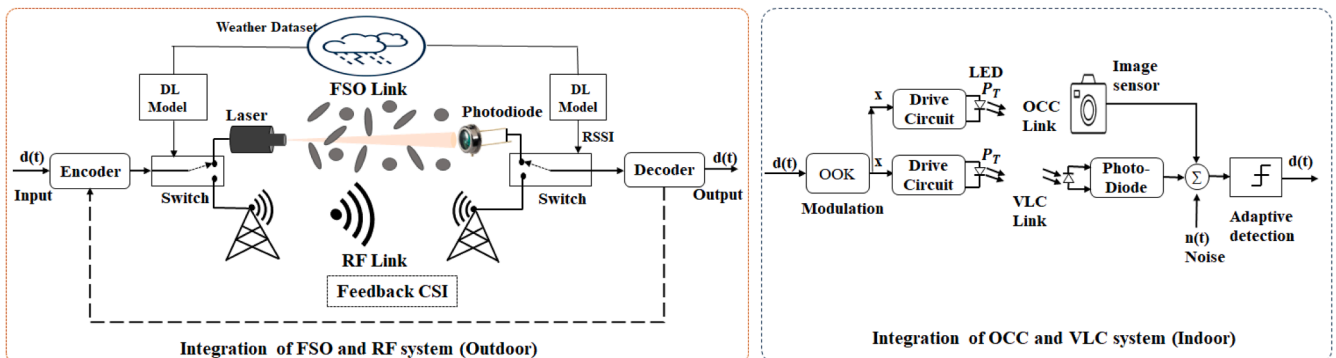


Fig. 2. System architecture of integration of OWC and RF.

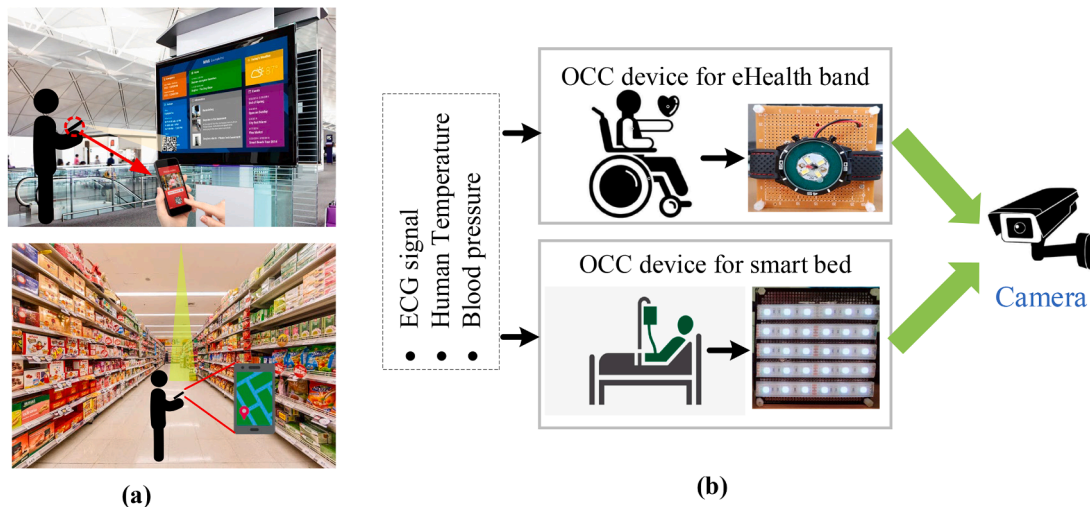


Fig. 3. Application scenarios of OWC for indoor environments.

systems, and support safe transportation as a 6 G standards requirement. Moreover, OWC can link a vast and varied range of 6 G equipment for both indoor/outdoor environments. The OWC system can enable extensive connectivity using LED for low power consumption, thus ensuring the IoT ecosystem remains applicable in the future. The OWC technologies should deploy for 6G/ beyond 6 G with many applications and coverage areas. For long communication (up to hundred km distance), FSO technology may apply to transmit high speed data system for Satellite to Satellite, building to building, Satellite to BS applications. For a short distance and the high data requirement, VLC/LiFi should use to take in advance the high bandwidth of light spectrum. OCC technologies may apply for harsh environment (raining, snowing, high mobility, etc.) with long distance (several hundred meter).

2.5. Enabling OWC technologies

In this section, we survey the progress of various OWC systems by considering four candidates: VLC, LiFi, OCC, and FSO. They offer promise for meeting the requirements of 6 G and beyond. The scenarios for OWC technologies for 6G/6 G beyond are depicted in Fig. 1. Fig. 2 shows the integration of OWC and RF technologies. As shown in Fig. 2, we may integration OWC technologies to improve communication system depending on advantages of VLC, LiFi, OCC, FSO to apply for types of applications, channel effect, and distances. With long communication up to hundred km distance, the FSO technology reveal as a good candidate to transmit high speed system, such as Satellite to Satellite, Satellite to BS, etc. With some challenging channel conditions, OCC technologies can resolve the harsh channel (raining, snowing, high mobility, etc.). With the short distance and the high data requirement, VLC/LiFi should use to take in advance the high bandwidth of light spectrum. Besides that, we may collaborate OWC and RF technologies. Since OWC and RF possess complementary characteristics, integrating them into one system is considered a viable approach to supporting next-generation communication systems (6 G and beyond 6 G). Hybrid RF/OWC systems offer a promising solution to mitigate the limitations of standalone systems while leveraging the strengths of both technologies.

2.6. - visible light communication (VLC)

VLC, one candidate of OWC, has emerged as a significant method. Its use of LED/PDs to transmit and receive optical signals. VLC is a promising candidate for the high-speed 6G/beyond 6 G systems. IoT systems are becoming increasingly important as they enable multiple devices to be connected for monitoring data and resource sharing data [51]. VLC

offers more than 10,000 times compared the RF system bandwidth [52]. By using the visible light spectrum, VLC offers opportunities in many applications, such as: smart factories, smart cars, and smart infrastructures, while being harmless to human health. However, the technology has some disadvantages: it is less effective for outdoor applications, sensitive with mobility environment, and short-range applications. Fig. 3(a) shows VLC localization applications for indoor environments.

LEDs can rapidly adjust their LED illumination, enabling data modulation at speeds undetectable by the human eye [53,54]. This allows energy-efficient LED devices to serve multiple functions, such as illumination, communication, and localization—capabilities that are unavailable in other technologies [55]. In [56], authors have demonstrated potential VLC data rates up to 10 Gbps with LEDs, and a mono-color LED has attained data rate of up to 3 Gbps. A significant disadvantage of using LEDs for transmitting data is the trade-off between bandwidth and optical efficiency. VLC can achieve up to 100 Gbps throughput for indoor lighting environments [57] and has some significant applications [58] as follows:

- Smart Lighting: VLC technology enables for illumination, communication, and light control, the it makes VLC system to be cheap and power consumption.
- V2X system: Due to numerous vehicles equipped with backlight/taillight, illumination equipments inside the car, and traffic lights utilizing LED technologies, VLC offers a excellent communication candidate for high secure, high-speed communication for V2V, V2I, and V2X systems.
- Hospitals: RF systems are often inappropriate in medical environments due to the impact on human health. VLC serves as a viable alternative to RF communications in these settings.
- Spectrum Relief: As RF bands become increasingly congested, VLC allows data offloading and can alleviate spectrum pressure.
- Underwater Communications: Unlike the RF system, which fails in underwater environments, VLC can provide a high data rate in aquatic environments.
- Smart Display Signboards: LED arrays are widely utilized for signage in locations such as malls, roads, airports, and bus stops, so data can be transmitted via VLC.

2.7. Optical camera communication

Smart items are increasingly prevalent in daily life, with most applications involving front and rear cameras that include flashlights. This

allows for the implementation of in using devices that utilize these flashes and cameras as transceivers without requiring additional hardware modifications. Therefore, many people use LED, LCD, or other smart lights using transmitters, with cameras or IS serving as receivers. The main differences between OCC and VLC are as follows. VLC specifically requires the VL spectrum for its signal, whereas OCC can utilize IR/VL/UV in VLC; either PDs or cameras/IS can function as receivers, but OCC exclusively uses cameras or IS. While extensive research has been conducted on indoor VLC systems, addressing the challenges of outdoor VLC remains critical. These challenges include limited communication range, significant signal attenuation, and interference from both artificial and natural light sources. Additionally, noise from ambient light, such as sunlight and streetlights, further complicates reliable signal detection. Overcoming these obstacles is essential for effectively implementing and expanding VLC/LiFi systems in outdoor environments, paving the way for the broader adoption and integration of this technology in real-world scenarios. Many studies have explored IS-based optical communication. For example, one study employed a CMOS machine for OCC, achieving data rates of up to 15 Mb/s and LED detection in real time [59]. Notably, the communication distance of an OCC system can be increased using a focal lens in the camera. A camera, effectively a two-dimensional array of PDs, typically contains over 0.01 billion pixels, each measuring 1–10 μm [60,61]. This high spatial resolution enables the camera to direct light from slightly different angles onto various sensor positions, supporting spatial-division multiplexing and MIMO systems in VLC. OCC has great potential for V2V, V2I, and I2I systems and may facilitate information sharing among mobile users. Figure 3(b) shows that the OCC is a promising technology for eHealth applications. By collecting health information based on health sensors, smart band and smart bed can transmit health data by LED. The signal can be collected by existing image sensors (CCTV, webcam, etc.) in hospital. Additionally, OCC can be used to detect and recognize objects, allowing users to transmit information through mobility environments. Some key advantages, disadvantages, and key applications of OCC technologies include the following.

Advantages of OCC

- No interference: When using an IS at the receiver side, light from various LEDs can be generally separated on a picture. The huge pixels in the IS, which allow each pixel to output optical signals independently, prevent signal mixing and facilitate communication even in the presence of surrounding lights, such as streetlights [62].
- Higher SNR values: The OCC system delivers a very high SNR, eliminating unexpected light sources among multiple expected light sources.
- Simplified processing: unexpected LEDs can be excluded from the IS by the receiver through simple computer vision algorithms. This reduces the complexity on the receiver side [62].
- Stable over long communication distances: The light signal power is recorded using IS, which can accommodate different distances by controlling camera parameters. Although the LED image size decreases as the range increases, the light power in each pixel remains consistent as long as the captured LED is larger than the pixel size [63].

Disadvantages of OCC

- Only NLOS Links: OCC relies on line-of-sight (LOS) communication, meaning that any light obstruction, such as buildings, fog, or dense gases, will disrupt the communication links [64].
- Data Rate Limitation: OCC has lower data rates than VLC/LiFi due to the camera frame rates. Typical ISs operate at approximately 30 fps, which limits the data rate to half or less of the fps to comply with Nyquist frequency requirements. To increase data rates, a high frame-rate camera is required [65].
- Flickering Effect: Because the camera frame rate remains low, transmitting signals at small frequencies using visible light can result

in flickering that is perceptible to the human eye, making it unsuitable for illumination [66].

- *Key Applications of OCC*
- V2X Communication: Cameras in vehicles can integrate with visible light and near-infrared (NIR) OWC technologies to achieve OCC systems that enable both monitoring and data communication, making it highly appealing for autonomous vehicles.
- Indoor Positioning: OCC systems can facilitate precise indoor positioning. By assigning unique identification (ID) information to LEDs, the receiver sides can detect and track infrastructures, animals, and target objects in smart homes [67].

2.8. - light fidelity (LiFi)

LiFi can be considered an OWC technology [68]. While both LiFi and VLC are optical communication technologies, they differ in several fundamental ways. VLC is limited to the VL signal, whereas LiFi utilizes VL for the uplink and may deploy IR/UV for the downlink [68–70]. VLC systems have either unidirectional/bidirectional [71] communication applications, while LiFi systems are characteristically bidirectional communication. User mobility in real-time is vital for LiFi, but it is not required for VLC [72]. Moreover, while VLC supports multiple communication modes, including point-to-point (P2P) and point-to-multipoint (P2MP), LiFi requires configurations of multiuser communication [73]. Based on that, we can highlight the unique strengths and applications of each technology. The application environments are similar for both technologies: indoor settings typically facilitate simultaneous communication and illumination, whereas outdoor scenarios may involve both functions (street light, car lights, etc.) or solely communication (traffic light, car lights, etc.). Figure 3(a) shows examples that are applicable to both LiFi and VLC, depending on the uplink communication channels that are supported. Other examples of VLC usage include communications from V2V, V2I, and V2X. LiFi is a better option than WiFi for sensitive environments, including hospital rooms, nuclear power plants, and airplane cabins, because it has more benefits than WiFi, such as faster data rates, lower cost, easily accessible spectrum, and improved security measures. It cannot, however, provide long-distance communications and is less useful for outdoor applications.

2.9. - free space optics (FSO)

Free space optics (FSO) [74–89], a type of OWC, typically operates using the near-infrared (NIR) spectrum, but it may also deploy within VL and UV spectra. The use of IR allows for lower attenuation levels. FSO systems primarily utilize laser diodes for transmission instead of LED or normal light sources. The light beams produced by LD enable high-speed links between the transmitter side and the receiver sides. FSO technologies are considered to support high-speed transmission over long distances that can range up to several thousand kilometers [40]. The modulation process in FSO systems can employ methods such as the OOK modulation, PSK modulation, or OFDM modulation. To enhance the intensity of the laser beam, optical amplifiers may be utilized, followed by collection and refocusing through beam-forming optics before transmission [40]. While laser diodes (LDs) are the standard transmitting sources for free-space optics, many researchers have also used high-power light sources with beam collimators. The transmitting light source requires high power, with size, power, and beam quality, among other factors, playing critical roles in determining laser intensity and minimum divergence [40]. Recent FSO technology has achieved data rates comparable to those of fiber optics [90–97]. Due to deploying easily, FSO has enabled many applications, including backhaul network, medical application, streaming application, broadcasting applications with high security. Furthermore, FSO systems are also utilized for metropolitan area network (MAN) to metropolitan area network and local area network (LAN) to local area network connections,

Table 1
Comparison of various wireless communication systems [19].

	LiFi	VLC	OCC	FSO	RF
Transmitter	LED	LED	LED	Laser	Antenna
Receiver	Photodiode	Photodiode/Image sensor	Image sensor	Photodiode	Antenna
Spectrum	IR/VL/UV	VL	IR/VL/UV	IR/VL/UV	Radio waves
Data rate	Very high (Gbps)	High (Mbps)	Moderate (kbps to Mbps)	Very high (Tbps)	Variable (kbps to Tbps)
Path loss	Medium (high for NLOS)	Medium (high for NLOS)	Less	High	High
Security	High	High	High	High	Low
Environment	Indoor	Indoor	Indoor/Outdoor	Outdoor	Indoor/Outdoor
Distance	10 m	20 m	200 m	~ km	~ km
Applications	Indoor high-speed communication	IoT, Smart lighting	AR, Indoor localization	Satellite, building to building communication	6 G, 5 G, Wi-Fi, etc.
Limitations	Short distance and non-mobility support	Short distance and non-mobility support	Lower data rate than LiFi/VLC	Long distance and non-mobility support	High interference

building-to-building communication, underwater communication, and satellite to satellite. Although FSO offers many advantages as showing before, there are challenges of FSO related to connect reliability, especially over long distances. These challenges arise from its sensitivity to various factors, including weather (fog, rain, and clouds), atmospheric interference, and physical obstructions, such as birds and tall buildings, that can disrupt the signal beam [43]. Additionally, as illustrated in Table 1, there are distinct differences between LiFi, VLC, FSO, OCC, and RF. VLC exclusively utilizes visible light for data transmission, whereas LiFi, FSO, and OCC can deploy not only visible light but also IR and UV spectrums, then it can enhance their communication capabilities. RF system can be applied for longer distance compared LiFi/VL-C/OCC/FSO, however, RF system has some limitations such as: low security, very high interference, and also harmful for human health. Fig. 1 shows that FSO is a promising technology for satellite, UAV, and underwater environments.

3. IEEE 802.15.7 standard for optical wireless communication

3.1. IEEE 802.15.7-2011

IEEE 802.15.7 standard [15] describes three physical specification mode (PHY) types: PHY I (11.67 kbps–266.6 kbps), PHY II (1.25 Mbps–96 Mbps), and PHY III (12 Mbps–96 Mbps), which utilizes Color Shift Keying (CSK) modulation [15]. OOK and Variable Pulse Position Modulation (VPPM), modulation methods for PHY I and PHY II, are both

designed to perform well in dimming controlling. PHY specification modes include modulation schemes for encoding LED signals and decoding the received signal from PDs, DC balance coding to reduce flicker effect, and FEC coding to reduce bit error rate. This standard also resolves the optical noise effect from both natural light sources and artificial light sources, to make sure that the compatibility between LEDs and PDs. PHY I specification mode achieved for low-rate VLC technologies considering outdoor environments. PHY II specification mode is suitable for high-rate VLC technologies considering indoor environment, whereas PHY III focuses on P2P using multiple light sources in indoor environment. The FEC methods are deployed to handle both long and short data packets, with Reed-Solomon (RS) coding and Convolutional Codes (CC) coding employed to mitigate many interferences for outdoor applications. On the contrary, indoor applications apply RS coding and run-length limited (RLL) coding to enhance VLC performances. Optical clock rates are defined in three PHY specification modes based on requirements of VLC applications: PHY I works at 400 KHz, PHY II works at 120 MHz, and PHY III works at 24 MHz, to ensure the communication performances tailored to the specific needs of each application.

3.2. IEEE 802.15.7-2018

To achieve higher performance in OWC technologies, some modulation and coding methods are proposed to amend IEEE 802.15.7–2011 with OCC specification mode. The IEEE 802.15.7 m Task Group (TG7m) committee has dedicated effort to the PHY/MAC layers and has accepted

Table 2
IEEE 802.15.7 Standard.

PHY	Description	Modulation schemes	Optical Clock rate	Technology	Data rate	Mobility
I	IEEE 802.15.7–2011 [15]	• OOK • VPPM	200–400 kHz	VLC	~ hundred kbps	Non-support
II	IEEE 802.15.7–2011 [15]	• OOK • VPPM	3.75–120MHz	VLC	~ hundred Mbps	Non-support
III	IEEE 802.15.7–2011 [15]	• CSK	12–24 MHz	VLC	~ hundred Mbps	Non-support
IV	IEEE 802.15.7–2018 [16]	• UFSOOK • Twinkle VPPM • S2-PSK • HS-PSK • Offset-VPWM	0.01–10 kHz	OCC	~ several kbps	Non-support
V	IEEE 802.15.7–2018 [16]	• RS-FSK • C–OOK • CM-FSK • MPM	10–12.5 kHz	OCC	~ several bps	Non-support
VI	IEEE 802.15.7–2018 [16]	• A-QL • HA-QL • VTASC	10–30 Hz	OCC	~ hundred kbps	Non-support
VII	IEEE 802.15.7a-2024 [17]	• Invisible data-embedding • RS-OFDM • MIMO C–OOK • O–NOMA • MIMO–OOK • Hybrid modulation	1.5–40 kHz	OCC	~several Mbps	~10 km/h

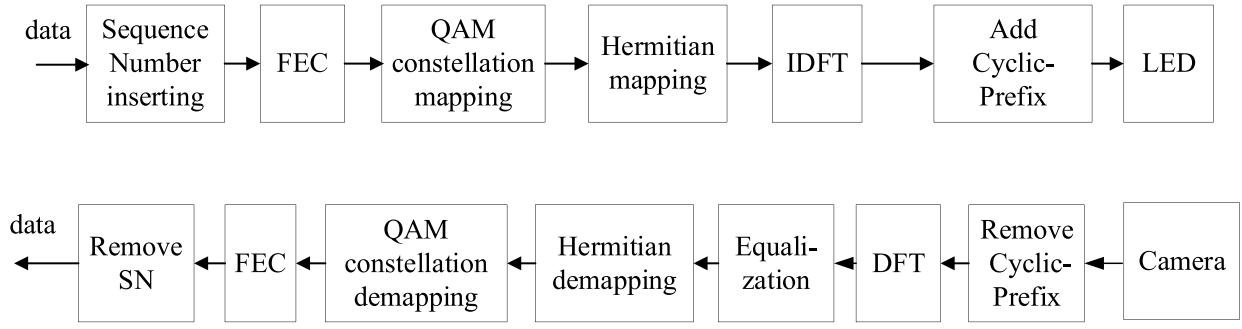


Fig. 4. System architecture of RS-OFDM scheme [18].

three new OCC PHY operating modes (PHY IV to PHY VI), as detailed in Table 2. While the IEEE 802.15.7–2011 standard is based on three fundamental modulations (OOK, VPPM, and CSK), TG7m introduces new OCC modes (PHY IV–VI) for camera-based applications. These modes operate at lower optical clock rates (below 100 kHz) and utilize baseband modulation, distinguishing them from the higher-rate PHY modes (VII and VIII) and traditional VLC/LiFi systems. This distinction presents unique technical challenges related to camera frame rates and shutter speeds. Moreover, although the newly proposed OCC modes are specifically designed for camera communication, PDs also have the potential to demodulate data from these modulated waveforms. In 2017, LiFi modes were reassigned to other task groups, namely the IEEE 802.15.13 MultiGigabit/s Optical Wireless Communications Task Group and the IEEE 802.11bb Task Group on Light Communication. These task group have been published with the following IEEE Std 802.11bb-2023 and IEEE Std 802.11bb-2023.

3.3. IEEE 802.15.7a-2024

Effective modulation and coding schemes are crucial to achieve high performance in OCC systems. Consequently, IEEE 802.15.7a-2024 (formerly known as TG7a) has made substantial investments in the PHY layer, which officially was published in February 2025. The committee approved a new PHY mode (PHY VII in Table 2) based on specification proposals from March 2022 through November 2024. On December 2024, with the recommendation of IEEE SA RevCom, the IEEE SASB approved to publish IEEE 802.15.7a-2024.

The latest TG7a draft document includes a variety of modulation techniques that emphasize multicarrier scheme approaches for OCC technologies. In contrast, the IEEE 802.15.7–2011 document utilized three fundamental methods (OOK, VPPM, and CSK [15], in PHY I, II, and

III in Table 2) for VLC technology, although the IEEE 802.15.7–2018 document implemented the low-speed OCC methods [16]. TG7a has already investigated a number of modulation strategies, such as OFDM, MIMO, and NOMA technologies, that are intended to improve data rate and communication distances. Furthermore, Deep Learning has been included to enhance OCC performance, especially in decoder operations.

3.4. OFDM technology

Inter-symbol interference (ISI) makes a lot of challenges in communication systems, for example: multipath and blur phenomena, which can make difficulties to enhance communication speed. Researchers employ OFDM rather than another fundamental modulation to reduce ISI. In high-speed communication systems, OFDM is a digital modulation that reduces the channel-distorting phenomenon of ISI by using multiple carrier frequencies. Compared to global shutter cameras, rolling-shutter cameras provide OCC technologies certain advantages since they allow pixel lines to be exposed and receive optical data with a high sampling rate. In OCC based rolling shutter effect systems, the sampling rate often exceed the selected optical clock rate by several kilohertz, fulfilling the Nyquist criterion and enhancing performance. Besides that, rolling shutter effect can easily deploy with OOK schemes. However, applying the OFDM technique to intensity modulation/direct detection (IM/DD) has a lot of challenges. PHY VII specifications in Table 2 present the use of the rolling-shutter effect for OFDM technology, called to RS-OFDM. This approach takes advantages from OFDM waveform and rolling shutter effect in OCC systems.

Fig. 4 shows the system architecture of the RS-OFDM modulation scheme with some differences compared OFDM based-RF. By using the rolling shutter effect to transmit the OFDM waveform, we can exploit the advantages of OFDM waveforms for the OCC system [98–100]. Unlike

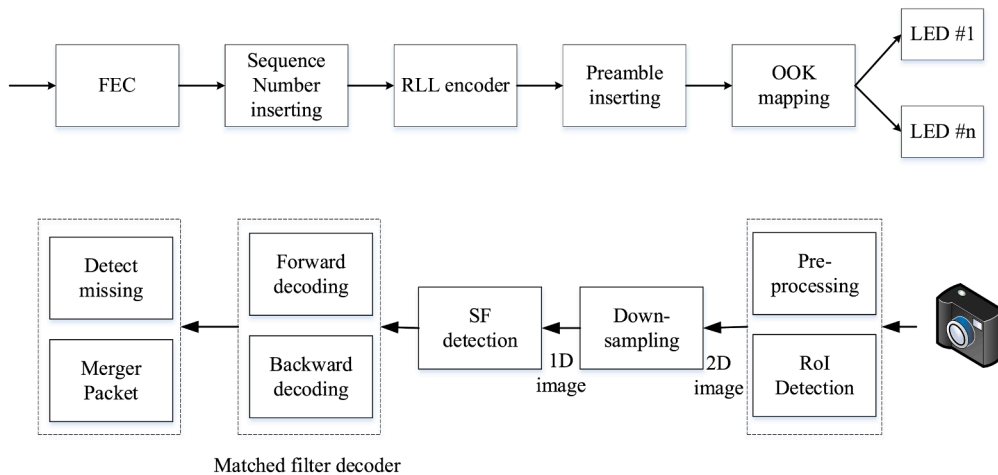


Fig. 5. System architecture of MIMO-COOK technique [18].

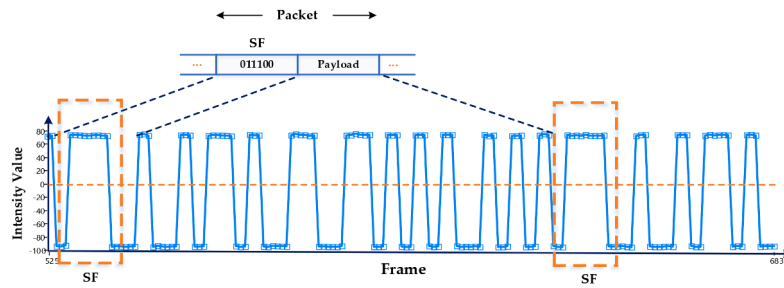


Fig. 6. Intensity of expected signals via RoI signaling [18].

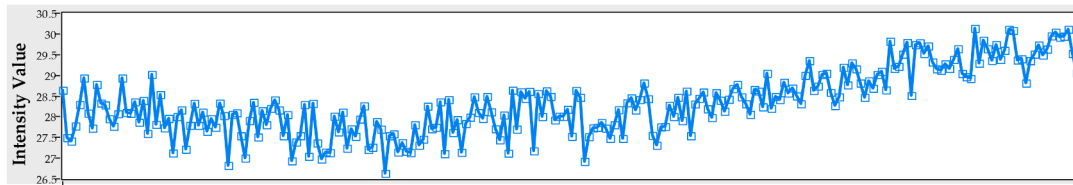


Fig. 7. Intensity of noise signals via RoI signaling [18].

traditional OFDM in RF, where data symbols are fed straight to IDFT, here, each symbol must first pass through a Hermitian block. This block ensures that the IDFT output remains fully real before the signal is sent to the IFFT.

3.5. MIMO technology

When using multiple-input multiple-output techniques, OCC systems have an obvious benefit over VLC and LiFi, successfully resolving the difficulties related to process many links with multiple LEDs. By using image processing methods, OCC can process with multiple LEDs simultaneously. RoI algorithms have been proposed by the IEEE 802.15.7–2018 standard to identify and monitor LED light sources. However, the deep learning approaches is enhancing object detection and tracking for faster processing and higher accuracy in mobility environment. It will improve MIMO performance in OCC systems [101–103]. Within the TG7a meetings, utilizing MIMO technology not only enhances data rates but also increases user capacity, overcoming limitations of the IEEE 802.15.7–2018 standard. This advancement highlights the potential of OCC systems to improve communication efficiency and performance in mobile environments.

Fig. 5 presents the architecture of the MIMO–COOK technique, as planned by TG7a. Unlike the conventional COOK in IEEE 802.15.7–2018, which utilizes RoI algorithms to detect light sources, we proposed a matched filter to decode signals and detect light sources. To detect preambles, we used a matched filter instead of the zero-crossing

filter, which revealed better performance than conventional works. In a real environment, the camera can distinguish between the signal from the expected LED and any unexpected light (sunlight or noise). The payload was inserted with preambles to easily predict the start of the frame, based on the expected signal. However, as shown in Figs. 6–7, we can identify the unexpected signal based on a simple approach (unexpected or noise signals have little variation intensity compared to expected light sources). Then we can efficiently differentiate the expected signal from noise using a simple approach.

3.6. O–NOMA technology

NOMA has emerged as a promising approach for OWC technologies. We can classify NOMA technology into code domain NOMA (CD-NOMA) and power domain NOMA (PD-NOMA) [104–106]. Unlike traditional methods that allocate frequency/time slots among users, PD-NOMA employs superposition coding to share transmitting power within the same time and frequency slots. Like to PD-NOMA, CD-NOMA deploys multiple users to utilize the same frequency and time slots through techniques like non-orthogonal low cross-correlation. At TG7a, O–NOMA was introduced in PHY VII, highlighting NOMA potential to enhance network capacity and efficiency in OCC systems. Fig. 8 shows the system architecture of the PD-NOMA technique for the OCC system. Noticeably, we can support multiple users using a single LED through power allocation. At the receiver side, the SIC should be applied to separate the power domain.

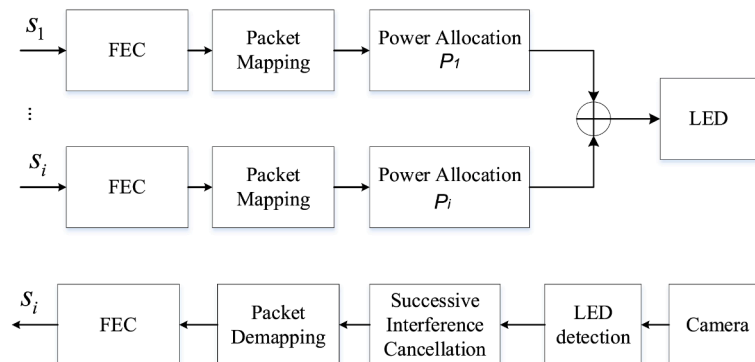


Fig. 8. System architecture of NOMA technique [18].

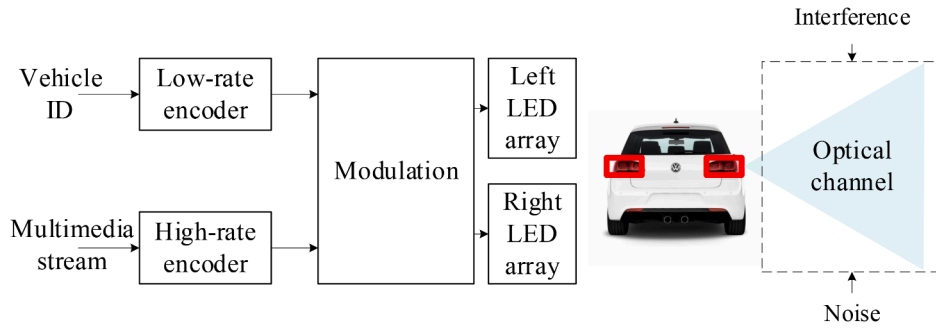


Fig. 9. System architecture of Hybrid modulation scheme [18].

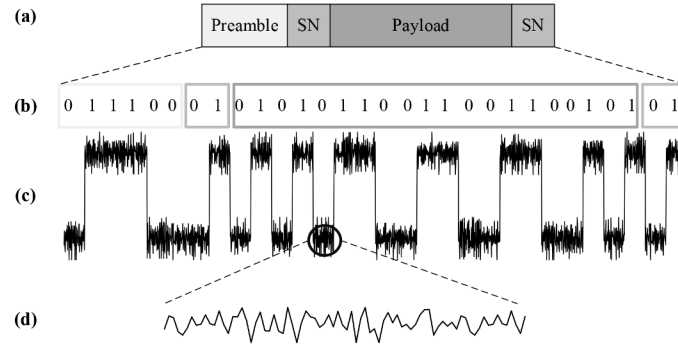


Fig. 10. Data frame structure of HOOK-OFDM scheme. (a) OOK packet. (b) OOK data (c) hybrid signal (d) OFDM signal [18].

3.7. Hybrid technology

Hybrid OCC technologies were designed to simultaneously transfer two signals (one for a high data rate and one for a low data rate) with a single light source. This approach can lower setup costs for OCC while offering multiple facilities to various users through a low-complexity setup. Moreover, the hybrid scheme saves energy by reducing the number of LEDs needed for the OCC system. From these benefits, hybrid modulation technologies were introduced in TG7a and considered under the PHY VII operation mode, which outlined in Table 2.

Fig. 9 shows the system architecture of a hybrid modulation scheme, which was proposed in TG7a. Instead of transmitting a single signal using a single LED, the hybrid modulation scheme will transmit multiple

signals in the same LED to save the cost. In TG7a, the hybrid modulation finds vehicular application. By using the existing LED in backlight/taillight, we can simultaneously transmit Vehicle ID and multimedia data, which is useful for vehicular applications. Fig. 10 shows the example of a HOOK-OFDM scheme, which implements PHY mode VII of IEEE 802.15.7a-2024, as listed in Table 2. With HOOK-OFDM, we can simultaneously transmit two signals, OOK and OFDM, with a single LED. We can then control the budget to design an OCC system. IEEE 802.15.7–2011 relied on basic modulation schemes (OOK, VPPM, and CSK) for VLC system, using only visible light spectrum for data transmission, unlike IEEE 802.15.7–2018 and IEEE 802.15.7a-2024, which also incorporate IR, VR, and UV spectrum. Furthermore, IEEE 802.15.7–2018 primarily supports the low rate, short range OCC in

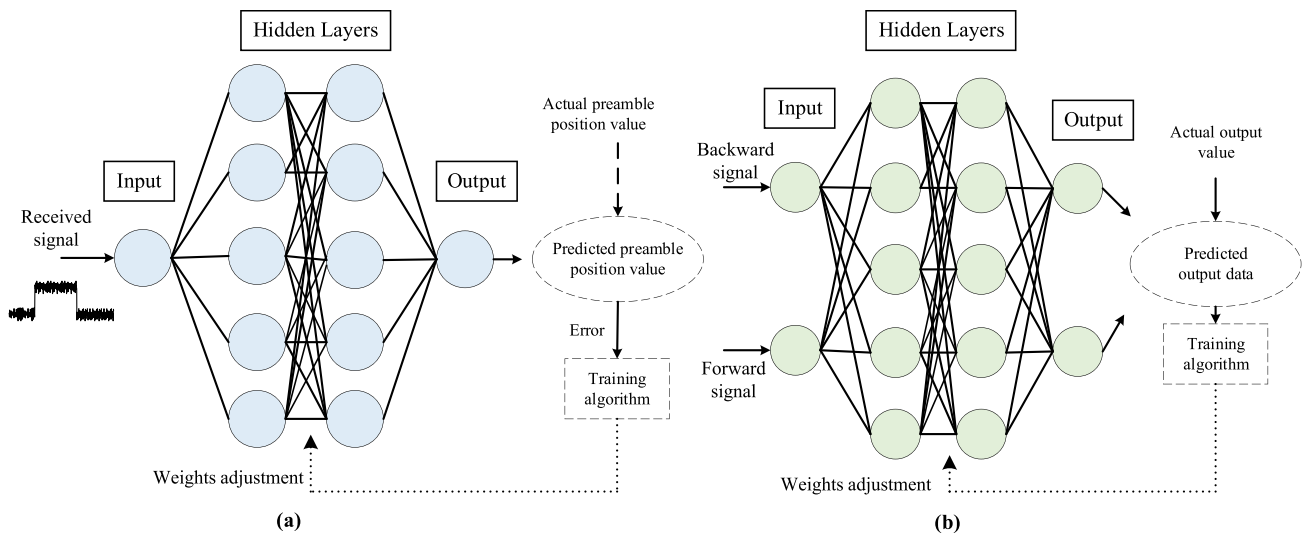


Fig. 11. Deep learning model for: (a) preamble detection, (b) decoding data.

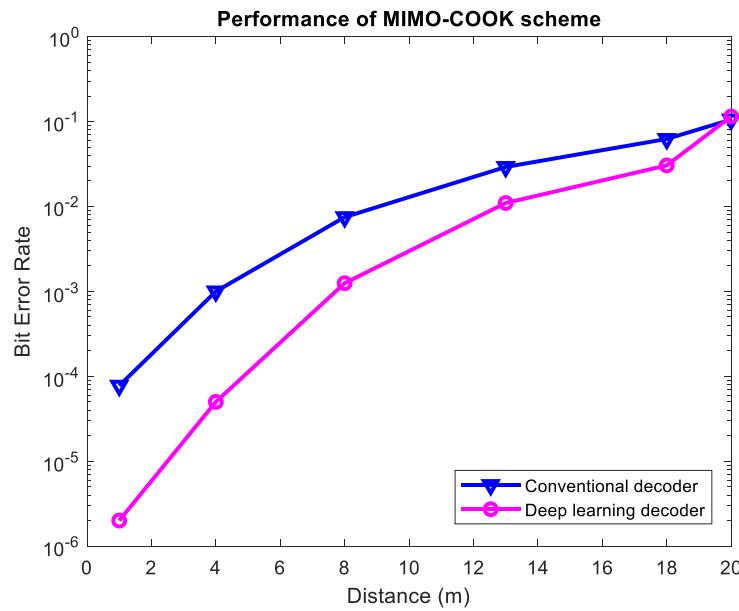


Fig. 12. BER performance of MIMO-COOK schemes with different distances considering a velocity of 2 m/s.

static environments, while IEEE 802.15.7a-2024 focuses on the higher rate, longer range OCC schemes considering mobility environments. As shown in Table 2, the mobility is achieved in IEEE 802.15.7a-2024 standard through the use of Deep Learning to detect the light sources and decode data. Also, the data rate of OCC is improved in IEEE 802.15.7a-2024 by integrating advanced technologies such as OFDM, MIMO, and NOMA. This paves the way for easier commercialization of OWC technologies for 6 G and beyond in the near future.

3.8. Deep learning decoder

In the IEEE 802.15.7–2018 standard, RoI algorithms are introduced as a potential solution for LED detection. However, they have proven ineffective in high-movement situations and when distinguishing between multiple light sources (both transmitting and non-transmitting). Next-generation OCC systems will require new detection technologies to tackle these problems. Recently, advancements in manufacturing have made deep learning a viable option for high-precision and reliable real-time object detection and tracking. Research has demonstrated that using You-Only-Look-Once in OCC receivers enhances the data speed associated with traditional methods. Beyond object detection, deep learning may also aid in data decoding to boost the OCC performance. When utilizing the rolling shutter effect to decode data, the striped patterns may overlap because of mobility effects or interference in other light (especially in MIMO setups) and long-range environments. Its factors can distort the OCC waveform and lead to decoding errors. Studies have indicated that implementing the DL model for decoding data can increase the OCC speed and reduce BER [107–110]. Fig. 11 shows the example of a deep learning model for preamble prediction. In the OCC system, the sampling process and the frame rate are randomly set, similar to in a real environment. Accordingly, and the preamble prediction becomes crucial for synchronizing the start of the frame. By applying the deep learning model for preamble prediction, we can reduce the bit error rate compared to conventional technologies. Additionally, Fig. 11 shows the deep learning model for data decoding. After predicting the preamble, the data has to be decoded to convert the signal to bit. For short-distance and non-harsh environments, we can easily decode the data due to the good SNR value. However, the SNR value will be lower if the distance between the transmitter and the receiver is long and the environment is harsh (raining, foggy, turbulent, etc.). Accordingly, deep learning should be considered to improve the OCC system.

Fig. 12 shows the bit error rate values at different distance considering mobility environment. As illustrated, using Deep Learning for preamble detection and data decoding significantly improves performance compared to conventional decoders in MIMO-COOK scheme. By applying Deep Learning, we may achieve BER of 10^{-4} at 6 m distance instead of 1 m distance with conventional decoder.

3.9. IG NG OWC

Automobiles are increasingly emerging as a key technology of the future, attracting significant attention from researchers and companies aiming to improve vehicle communication and safety systems. As many automotive manufacturers process to improve safety features beyond unmanned vehicles, the development of an OWC system for vehicular environments becomes essential for vehicle communication. Autonomous vehicles rely on multi-sensor, such as LiDAR, RADAR, camera, ultrasonic, and infrared, each sensors supporting each others to robust data collection framework. However, developing an ultra-reliable autonomous vehicular system remains challenges in real-time processing. While high-speed trains, such as Korea's next-generation KTX (HEMU-430X) and CRRC Maglev train in China, can achieve speeds exceeding 400 km/h. However, the current TG7a standards are limited to lower speeds of 200 km/h and distances of 200 m. From that, we need a new standard that address high-speed, long-range OWC to better support the current autonomous vehicular technology. The interest group next-generation OWC (IG NG OWC) was created in January 2024 to propose a new approach for the next-generation OWC for high rates (up to 200 Mb/s) and longer range (up to ~km) to maintain connectivity in high-speed environments (speeds of up to 400 km/h) and localization levels below 1 cm. To resolve these targets, some technologies have been proposed, such as laser communication, joint communications and sensing (LiDAR, RADAR, etc.), high-mobility 3D communication applications, bidirectional communication, MAC protocol for full duplex operation, and quantum technology. The details are explained in Section IV.

Table 3

Summary of IEEE 802.15.7a-2024 [14] and next generation OCC.

	IEEE 802.15.7a-2024	Next-generation OCC
PHY layer	Unidirectional communication Only communication Ordinary IR/VL/UV Light	Bidirectional communication Joint communications and sensing IR/VL/UV Laser
MAC layer	Unslotted ALOHA and CSMA/CA	AI based-MAC protocols

4. Summary and future directions of OWC for 6G and beyond 6G systems

4.1. Summary

Wireless communication technologies possess unique characteristics, including different standardizations, modulation schemes, physical transmitters and receivers, operational principles, attenuation traits, and safety levels. VLC can serve multiple functions, such as illumination, communication, and localization, whereas LiFi focuses on brilliance and transfer information. VLC and LiFi may achieve high data rate, but they are ineffective in outdoor or long-range environments because they are sensitive to sunlight and noise light sources. OCC offers a good idea for supporting long-range communication, object detection, and positioning application for indoor and outdoor considering mobility environments. However, OCC have some main limitations, such as: the low data rate depending on the camera frame rate. FSO enables high-speed and ultra-long communication distances but can be affected by factors from optical channel conditions. LiDAR is particularly effective for high-resolution 3D mapping, and it is expected to support OWC systems for higher rate and longer range in the near future. Each OWC technology has its own application areas that need to be leveraged for optimal performance. OWC can achieve higher communication channel capacity compared to RF-based communications as well as enhanced security. However, OWC is susceptible to obstructions that can block the signal with NLOS environment. Therefore, integrating RF and OWC systems is a promising approach for the future of wireless communications, allowing the advantages of each technology to be effectively utilized.

4.2. Future directions

TG7a is making the basis for high-rate, long-range OWC compared the existing IEEE 802.15.7 standard, which focuses on visible light communication in 2011 and low rate, short range OCC in 2018. While the current TG7a specification favor unidirectional OCC, next generation OWC standards are expected to address limitations such as: the lack of bidirectional communication, which is crucial for enhanced reliability of OWC system.

The hybrid technologies for OWC and RF systems is a promising candidate for the development of next generation communication system that can overcome the limitations of standalone technologies, particularly in 6 G and beyond 6 G system [114]. Quantum Key Distribution (QKD) is promising technology for future OWC standards, particularly for security features in military applications. Future standards should be applied joint communication and sensing, which will be important for 6 G and beyond 6 G. The Intelligent Reflecting Surfaces to improve LiFi, FSO, and VLC performance is notable [115], although it has some challenges for OCC systems and understanding IRS impact for optical channel model. Table 3 presents several promising directions for

Table 5

Summary of Current status of FSO and next generation FSO.

	Current status of FSO	Next-generation FSO
PHY Layer	Data rate: 13.16 Tbps [113] High sensitive with environment Non-mobility support	Data rate: up to 100 Tbps Low sensitive with environment by joint communications and sensing Mobility support
MAC services	Non-AI support	AI based-MAC protocols

the next-generation OCC technologies to enhance the current TG7a in MAC and PHY layers, including bidirectional communication, joint communications and sensing, AI based-MAC protocol, and, notably, Laser sensor instead of ordinary light sources. Laser sensors have emerged as promising candidates for next-generation OCC systems capable of long-distance communication, but their sensitivity in mobile environments presents a significant challenge. The integration of deep learning for tracking and object detection could enhance laser OCC performance. This innovation, along with the development of a new MAC protocol to support bidirectional communication and full-duplex optical MAC (FD-OMAC) [116], is anticipated in future OWC standards beyond the TG7a, which is slated for completion in 2024.

Besides that, in Table 4 and 5, we proposed the future directions for high speed VLC, LiFi, and FSO by apply joint communications and sensing to reduce the sensitive effect with mobility channel conditions for VLC/LiFi/FSO technologies. To offer services to the upper protocol layers, IEEE 802.15.13 apply MD-SAP (MAC data service access point) and MLME-SAP (MAC layer management entity service access point) [111]. With IEEE 802.11bb-2023, HT (high-throughput), VHT (very high-throughput), or HE (high-efficiency) MAC procedures [112] support for Light Communication PHY layer. AI-based MAC layer promises to enhance the performance of 6 G systems [117] in general and the next generation VLC/LiFi/FSO in particular, including latency and energy consumption. Overall, while the current standards (IEEE 802.15.7, IEEE 802.15.13–2023, and IEEE 802.11bb-2023) represent many advantages compared previous standards, the commercialization and broader adoption of OWC systems should be considered for 6 G and beyond 6 G.

5. Conclusion

Existing RF technologies are inadequate to meet the quickly growing demand for wireless capacity in 6G/6 G beyond and future communications. The extensive optical spectrum, encompassing IR, VL, and UV spectrums, presents an opportunity for wireless connectivity for 6 G and IoT systems. Utilizing IR/VL/UV spectrums alongside RF can alleviate problems related to spectrum scarcity in the RF system. Many researchers are working to transition various RF communication applications to OWC, which is poised to significantly impact industries such as healthcare, public venues management, transportation, residential housing, shopping malls, underwater communication, and satellite applications. This survey presents a comprehensive overview of promising OWC systems, categorizing them from multiple perspectives. Key technologies discussed include VLC, LiFi, OCC, and FSO, which facilitate communication from ultra-short to ultra-long distances. Each technology has distinct advantages and limitations, leading to varied application areas. We highlight the differences in basic principle architectures

Table 4

Summary of IEEE 802.15.13–2023 [111], IEEE 802.11bb-2023 [112], and next generation VLC/LiFi.

	IEEE 802.15.13–2023	IEEE 802.11bb-2023	Next-generation VLC/LiFi
PHY Layer	Data rate: 2.192 Gbps Only communication Non-mobility support	Data rate: 9.6 Gbps Only communication Non-mobility support	Data rate: up to 100 Gbps Joint communications and sensing Mobility support
MAC Layer	MD-SAP and MLME-SAP based-MAC protocol	LC HT/HE/VHT MAC protocols	AI based-MAC protocols

and applications for these OWC technologies and outline the challenges they face. Overall, optical wireless technologies hold great promise for enhancing the quality of wireless communication systems. This comparative review can serve as a valuable resource for understanding advancements in promising OWC technologies while encouraging continued efforts toward integrating OWC systems as a vital complement to RF-based technologies in future 6 G and 6 G beyond heterogeneous wireless networks. Besides that, the next-generation OWC is proposed to enhance the current OWC standard at the MAC and PHY layers, including bidirectional communication, joint communications and sensing, AI based-MAC protocol, and, notably, laser OCC instead of ordinary light sources for high mobility environment to meet the requirements of 6 G and 6 G beyond system.

CRedit authorship contribution statement

Huy Nguyen: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Al-Imran:** Visualization, Formal analysis. **Yeong Min Jang:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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References

- [1] X. Chen, J. Tan, L. Kang, F. Tang, M. Zhao, N. Kato, Frequency Selective Surface Toward 6G Communication Systems: a Contemporary Survey, *IEEE Communications Surveys & Tutorials* 26 (3) (2024) 1635–1675, thirdquarter.
- [2] L. Jiao, et al., Advanced Deep Learning Models for 6G: overview, Opportunities, and Challenges, *IEEE Access* 12 (2024) 133245–133314.
- [3] W. Jiang, B. Han, M.A. Habibi, H.D. Schotten, The Road Towards 6G: a Comprehensive Survey, *IEEE Open Journal of the Communications Society* 2 (2021) 334–366.
- [4] M.A. Ferrag, et al., Edge Learning for 6G-Enabled Internet of Things: a Comprehensive Survey of Vulnerabilities, Datasets, and Defenses, *IEEE Communications Surveys & Tutorials* 25 (4) (2023) 2654–2713, Fourthquarter.
- [5] N.P. Kuruvatti, M.A. Habibi, S. Partani, B. Han, A. Fellan, H.D. Schotten, Empowering 6G Communication Systems With Digital Twin Technology: a Comprehensive Survey, *IEEE Access* 10 (2022) 112158–112186.
- [6] U.M. Malik, M.A. Javed, S. Zeaddly, S.u. Islam, Energy-Efficient Fog Computing for 6G-Enabled Massive IoT: recent Trends and Future Opportunities, *IEEE Internet of Things Journal* 9 (16) (2022) 14572–14594, 15 Aug.15.
- [7] C.-X. Wang, et al., On the Road to 6G: visions, Requirements, Key Technologies, and Testbeds, *IEEE Communications Surveys & Tutorials* 25 (2) (2023) 905–974, Secondquarter.
- [8] D. Serghiou, M. Khalily, T.W.C. Brown, R. Tafazolli, Terahertz Channel Propagation Phenomena, Measurement Techniques and Modeling for 6G Wireless Communication Applications: a Survey, *Open Challenges and Future Research Directions*, *IEEE Communications Surveys & Tutorials* 24 (4) (2022) 1957–1996, Fourthquarter.
- [9] T.Q. Duong, D. Van Huynh, Y. Li, E. Garcia-Palacios, K. Sun, Digital Twin-enabled 6G Aerial Edge Computing with Ultra-Reliable and Low-Latency Communications: (Invited Paper), in: 2022 1st International Conference on 6G Networking (6GNet, Paris, France, 2022, pp. 1–5.
- [10] M. Yadav, et al., Exploring Synergy of Blockchain and 6G Network for Industrial Automation, *IEEE Access* 11 (2023) 137163–137187.
- [11] M. Rowshan, M. Qiu, Y. Xie, X. Gu, J. Yuan, Channel Coding Toward 6G: technical Overview and Outlook, *IEEE Open Journal of the Communications Society* 5 (2024) 2585–2685.
- [12] D. Tsonev, S. Videv, H. Haas, Towards a 100 Gb/s visible light wireless access network, *Opt. Exp.* 23 (2) (2015) 1627–1637, Jan.
- [13] L. Hanzo, H. Haas, S. Imre, D. O'Brien, M. Rupp, L. Gyongyosi, Wireless myths, realities, and futures: from 3G/4G to optical and quantum wireless, *Proc. IEEE* 100 (2012) 1853–1888, May.
- [14] H. Nguyen, I.B. Krishna, Yoga Utama, Y.M. Jang, Enabling Technologies and New Challenges, *IEEE 802.15.7 Optical Camera Communications Standard*, *IEEE Communications Magazine* 62 (3) (2024) 90–95, March.
- [15] IEEE Standard for Local and Metropolitan Area Networks — Part 15.7: short-Range Wireless Optical Communication Using Visible Light, *IEEE Standard 802 (15)* (2011) 7. -2011.
- [16] IEEE Standard for Local and Metropolitan Area Networks — Part 15.7: short-Range Wireless Optical Communication, *IEEE Standard 802 (15)* (2018) 7. -2018.
- [17] IEEE Mentor 802.15.7a Task Group docs; https://mentor.ieee.org/802.15/documents?is_group=007a, accessed November 11, 2023.
- [18] Huy Nguyen and Yeong Min Jang, Draft D2-Merged Document for IEEE 802.15.7a TG, [Online]. Available: <https://mentor.ieee.org/802.15/dcn/22/15-22-0332-00-007a-draft-d2-merged-document-for-ieee-802-15-7a-tg.docx>.
- [19] M.Z. Chowdhury, M.T. Hossain, A. Islam, Y.M. Jang, A Comparative Survey of Optical Wireless Technologies: architectures and Applications, *IEEE Access* 6 (2018) 9819–9840.
- [20] H. Chen, H.P.A. van den Boom, E. Tangdiongga, T. Koonen, 30-Gb/s bidirectional transparent optical transmission with an MMF access and an indoor optical wireless link, *IEEE Photon. Technol. Lett.* 24 (7) (2012) 572–574, Apr. 1.
- [21] D.K. Borah, A.C. Boucouvalas, C.C. Davis, S. Hranilovic, K. Yiannopoulos, A review of communication-oriented optical wireless systems, *EURASIP J. Wireless Commun. Netw.* 2012 (2012) 91, Dec.
- [22] S. Dimitrov, H. Haas, Information rate of OFDM-based optical wireless communication systems with nonlinear distortion, *J. Lightw. Technol.* 31 (6) (2013) 918–929, Mar. 15.
- [23] H. Elgala, R. Mesleh, H. Haas, Indoor optical wireless communication: potential and state-of-the-art, *IEEE Commun. Mag.* 49 (9) (2011) 56–62, Sep.
- [24] Z. Ghassemlooy, P. Luo, S. Zvanovec, Optical camera communications. *Optical Wireless Communications*, Springer, Cham, Switzerland, 2016, pp. 547–568, Aug.
- [25] J.B. Carruthers, *Wireless infrared communications*. Wiley Encyclopedia of Telecommunications, Wiley, New York, NY, USA, 2003.
- [26] Z. Xu, B.M. Sadler, Ultraviolet communications: potential and state-of-the-art, *IEEE Commun. Mag.* 46 (5) (2008) 67–73, May.
- [27] Z. Xu, G. Chen, F. Abou-Galala, M. Leonardi, Experimental performance evaluation of non-line-of-sight ultraviolet communication systems, in: *Proc. SPIE* 6709, 2007 67090Y, Aug.
- [28] H. Qin, Y. Zuo, D. Zhang, Y. Li, J. Wu, Received response based heuristic LDPC code for short-range non-line-of-sight ultraviolet communication, *Opt. Exp.* 25 (5) (2017) 5018–5030, Mar.
- [29] P.H. Pathak, X. Feng, P. Hu, P. Mohapatra, Visible light communication, networking, and sensing: a survey, potential and challenges, *IEEE Commun. Surveys Tuts.* 17 (4) (2015) 2047–2077, 4th Quart.
- [30] F. Yang, J. Gao, Dimming control scheme with high power and spectrum efficiency for visible light communications, *IEEE Photon. J.* 9 (1) (2017) 7901612, FebArt.
- [31] Y.-J. Zhu, Z.-G. Sun, J.-K. Zhang, Y.-Y. Zhang, J. Zhang, Training receivers for repetition-coded MISO outdoor visible light communications, *IEEE Trans. Veh. Technol.* 66 (1) (2017) 529–540, Jan.
- [32] S. Dimitrov, H. Haas, *Principles of LED Light Communications: Towards Networked Li-Fi*, Cambridge Univ. Press, Cambridge, U.K., 2015, Mar.
- [33] H.-H. Lu, et al., A 56 Gb/s PAM4 VCSEL-based LiFi transmission with two-stage injection-locked technique, *IEEE Photon. J.* 9 (1) (2017) 7900208, FebArt.
- [34] H. Haas, L. Yin, Y. Wang, C. Chen, What is LiFi? *J. Lightw. Technol.* 34 (6) (2016) 1533–1544, Mar. 15.
- [35] Y.-C. Chi, et al., Phosphorous diffuser diverged blue laser diode for indoor lighting and communication, *Sci. Rep.* 5 (2015) 18690, Dec.
- [36] D. Tsonev, Brien, et al., A 3-Gb/s single-LED OFDM-based wireless VLC link using a gallium nitride μ LED, *IEEE Photon. Technol. Lett.* 26 (7) (2014) 637–640, Apr. 1.
- [37] Y. Goto, et al., A new automotive VLC system using optical communication image sensor, *IEEE Photon. J.* 8 (3) (2016) 6802716, Jun.
- [38] T. Yamazato, et al., Image-sensor-based visible light communication for automotive applications, *IEEE Commun. Mag.* 52 (7) (2014) 88–97, Jul.
- [39] Z. Zheng, L. Liu, W. Hu, Accuracy of ranging based on DMT visible light communication for indoor positioning, *IEEE Photon. Technol. Lett.* 29 (8) (2017) 679–682, Apr. 15.
- [40] M.A. Khalighi, M. Uysal, Survey on free space optical communication: a communication theory perspective, *IEEE Commun. Surveys Tuts.* 16 (4) (2014) 2231–2258, 4th Quart.
- [41] W.-S. Tsai, et al., A 20-m/40-Gb/s 1550-nm DFB LD-based FSO link, *IEEE Photon. J.* 7 (6) (Dec. 2015) 7905907.
- [42] D.L. Begley, Free-space laser communications: a historical perspective, in: *Proc. Annu. Meeting, IEEE Lasers Electro-Opt. Soc.*, 2002, pp. 391–392, Nov.
- [43] A. Malik, P. Singh, Free space optics: current applications and future challenges, *Int. J. Opt.* 2015 (2015) 945483, Sep.
- [44] J.C. Juarez, A. Dwivedi, A.R. Hammons, S.D. Jones, V. Weerackody, R.A. Nichols, Free-space optical communications for next-generation military networks, *IEEE Commun. Mag.* 44 (11) (2006) 46–51, Nov.
- [45] T. Nguyen, M. Hossain, Y. Jang, Design and implementation of a novel compatible encoding scheme in the time domain for image sensor communication, *MDPI Sensors* 16 (2016) 736.
- [46] Huy Nguyen, Minh Duc Thieu, Trang Nguyen, Yeong Min Jang, Rolling OFDM for Image Sensor based Optical Wireless Communication, *IEEE Photonics J.* (2019).
- [47] Minh Duc Thieu Van Hoa Nguyen, Huy Nguyen, Yeong Min Jang, “Design and Implementation of the MIMO-COOK Scheme Using an Image Sensor For Long-Range Communication, *MDPI Sensors*, 2020.
- [48] M.Z. Chowdhury, et al., A Comparative Survey of Optical Wireless Technologies: architectures and Applications, *IEEE Access* 6 (2018) 220, 9819–10.

- [49] S. Gopi, S. Kalyani, L. Hanzo, Cooperative 3D Beamforming for Small-Cell and Cell-Free 6G Systems, *IEEE Transactions on Vehicular Technology* 71 (5) (2022) 5023–5036. May.
- [50] C.-W. Chow, “Recent Advances and Future Perspectives in Optical Wireless Communication, Free Space Optical Communication and Sensing for 6G,” *Journal of Lightwave Technology*, vol. 42, no. 11, pp. 3972–3980.
- [51] R. Boubezari, H.L. Minh, Z. Ghassemloooy, A. Bouridane, Smartphone camera based visible light communication, *J. Lightw. Technol.* 34 (17) (2016) 4120–4126. Sep. 1.
- [52] D. Karunatilaka, F. Zafar, V. Kalavally, R. Parthiban, LED based indoor visible light communications: state of the art, *IEEE Commun. Surveys Tuts.* 17 (3) (2015) 1649–1678, 3rd Quart.
- [53] T. Komine, M. Nakagawa, Fundamental analysis for visible light communication system using LED lights, *IEEE Trans. Consum. Electron.* 50 (1) (2004) 100–107. Feb.
- [54] D.C. O’Brien, L. Zeng, H. Le-Minh, G. Faulkner, J.W. Walewski, S. Randel, Visible light communications: challenges and possibilities, in: *Proc. Int. Symp. Pers., Indoor Mobile Radio Commun, Cannes, France*, 2008, pp. 1–5. Sep.
- [55] H. Liu, H. Darabi, P. Banerjee, J. Liu, Survey of wireless indoor positioning techniques and systems, *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.* 37 (6) (2007) 1067–1080. Nov.
- [56] F. Zafar, M. Bakaul, R. Parthiban, Laser-diode-based visible light communication: toward gigabit class communication, *IEEE Commun. Mag.* 55 (2) (2017) 144–151. Feb.
- [57] L.U. Khan, Visible light communication: applications, architecture, standardization and research challenges, *Digit. Commun. Netw.* 3 (2) (2016) 78–88. May.
- [58] I. Takai, S. Ito, K. Yasutomi, K. Kagawa, M. Andoh, S. Kawahito, LED and CMOS Image Sensor Based Optical Wireless Communication System for Automotive Applications, *IEEE Photonics J* 5 (5) (2013) 6801418. -6801418Oct.
- [59] *The Canon EOS-1Ds MARK II: The Absolute Pinnacle of d-SLR Design and Performance*, Canon, Tokyo, Japan, 2004.
- [60] M. Schöberl, A. Brückner, S. Foessel, A. Kaup, Photometric limits for digital camera systems, *J. Electron. Imag.* 21 (2) (2012) 020501–020503. Jun.
- [61] I. Takai, S. Ito, K. Yasutomi, K. Kagawa, M. Andoh, S. Kawahito, LED and CMOS image sensor based optical wireless communication system for automotive applications, *IEEE Photon. J.* 5 (5) (2013) 6801418. OctArt.
- [62] Huy Nguyen, Nam Tuan Le, Duy Tuan Anh Le, Yeong Min Jang, “Design and Implementation of a Hybrid Optical Camera Communication System for Indoor Applications,” *MDPI Sensors*, 2024.
- [63] Huy Nguyen, Linh Van, Duc Nguyen, Yeong Hoang Tran, Min Jang, Rolling Shutter OFDM scheme for Optical Camera Communication considering Mobility Environment based on Deep Learning, *MDPI Applied Sciences* (2022).
- [64] W.A. Cahyadi, et al., Optical Camera Communications: principles, Modulations, Potential and Challenges, *Electronics (Basel)* 9 (2020) 1339.
- [65] Huy Nguyen, Van Hoa Nguyen, Cong Hoan Nguyen, Van Bui, Yeong Min Jang, “Design and Implementation of 2D MIMO-Based Optical Camera Communication Using a Light-Emitting Diode Array for Long-Range Monitoring System,” *MDPI Sensors*, 2021.
- [66] H. Shimizu, et al., Experimental Demonstration of Optical OFDM With Polarity-Separated Transmission in Rolling Shutter Based Visible Light Communication, in: *2023 IEEE International Symposium on Circuits and Systems (ISCAS)*, Monterey, CA, USA, 2023, pp. 1–5.
- [67] Minh Duc Thieu Van Hoa Nguyen, Huy Nguyen, Yeong Min Jang “Design and Implementation of the MIMO-COOK Scheme Using an Image Sensor For Long-Range Communication, *MDPI Sensors*, 2020.
- [68] M.A. Arfaoui, et al., Measurements-Based Channel Models for Indoor LiFi Systems, *IEEE Transactions on Wireless Communications* 20 (2) (2021) 827–842. Feb.
- [69] Y. Huang, M. Safari, H. Haas, I. Tavakkolnia, Optical Wireless 3-D-Positioning and Device Orientation Estimation, *IEEE Open Journal of the Communications Society* 5 (2024) 4519–4530.
- [70] M. Ayyash, et al., Coexistence of WiFi and LiFi toward 5G: concepts, opportunities, and challenges, *IEEE Commun. Mag.* 54 (2) (2016) 64–71. Feb.
- [71] S. Li, A. Pandharipande, F.M.J. Willems, Unidirectional visible light communication and illumination with LEDs, *IEEE Sensors J* 16 (23) (2016) 8617–8626. Dec.
- [72] W. Yuanquan, C. Nan, A high-speed bi-directional visible light communication system based on RGB-LED, *China Commun* 11 (9) (2014) 40–44. Mar.
- [73] H. Haas, High-speed wireless networking using visible light, *SPIE Newsroom* 1 (2013) 1–3. Apr.
- [74] M.Z. Chowdhury, M.K. Hasan, M. Shahjalal, M.T. Hossain, Y.M. Jang, Optical Wireless Hybrid Networks: trends, Opportunities, Challenges, and Research Directions, *IEEE Communications Surveys & Tutorials* 22 (2) (2020) 930–966. Secondquarter.
- [75] A.A. El-Malek, A.M. Salhab, S.A. Zummo, M.-S. Alouini, Effect of RF interference on the security-reliability tradeoff analysis of multiuser mixed RF/FSO relay networks with power allocation, *J. Lightw. Technol.* 35 (9) (2017) 1490–1505. May 1.
- [76] H. Yuksel, S. Milner, C. Davis, Aperture averaging for optimizing receiver design and system performance on free-space optical communication links, *J. Opt. Netw.* 4 (8) (2005) 462–475.
- [77] M.K. Hasan, M.O. Ali, M.H. Rahman, M.Z. Chowdhury, Y.M. Jang, Optical Camera Communication in Vehicular Applications: a Review, *IEEE Transactions on Intelligent Transportation Systems* 23 (7) (2022) 6260–6281. July.
- [78] B. Smutny, et al., 5.6 Gbps optical intersatellite communication link, in: *Proc. SPIE* 7199, . 2009 719906. Feb.
- [79] D. Killinger, Free space optics for laser communication through the air, *Opt. Photon. News* 13 (10) (2002) 36–42. Oct.
- [80] H. Henniger, O. Wilfert, An introduction to free-space optical communications, *Radio engineering* 19 (2) (2010) 203–212. Jun.
- [81] X. Liu, Free-space optics optimization models for building sway and atmospheric interference using variable wavelength, *IEEE Trans. Commun.* 57 (2) (2009) 492–498. Feb.
- [82] M.-C. Jeung, et al., 8×10 Gb/s terrestrial optical free space transmission over 3.4 km using an optical repeater, *IEEE Photon. Technol. Lett.* 15 (1) (2003) 171–173. Jan.
- [83] E. Ciaramella, et al., 1.28-Tb/s (32 × 40 Gb/s) free-space optical WDM transmission system, *IEEE Photon. Technol. Lett.* 21 (16) (2009) 1121–1123. Aug. 15.
- [84] Y. Kaymak, R. Rojas-Cessa, J. Feng, N. Ansari, M. Zhou, On divergence-angle efficiency of a laser beam in free-space optical communications for high-speed trains, *IEEE Trans. Veh. Technol.* 66 (9) (2017) 7677–7687. Sep.
- [85] A.K. Majumdar, J.C. Ricklin, *Free-Space Laser Communications: Principles and Advances*, Springer-Verlag, New York, NY, USA, . 2010. Dec.
- [86] S.S. Muhammad, B. Flecker, E. Leitgeb, M. Gebhart, Characterization of fog attenuation in terrestrial free space optical links, *Opt. Eng.* 46 (6) (2007) 066001–066010. Jun.
- [87] M.L.B. Riediger, R. Schober, L. Lampe, Fast multiple-symbol detection for free-space optical communications, *IEEE Trans. Commun.* 57 (4) (2009) 1119–1128. Apr.
- [88] Al-Imran, M.S.Nazim, H. Nguyen, Y.M. Jang, “Channel estimation of massive MIMO FSO communication system using deep attention residual U-Net,” *ICT Express*, 2024.
- [89] H. Kaushal, G. Kaddoum, Optical communication in space: challenges and mitigation techniques, *IEEE Commun. Surveys Tuts.* 19 (1) (2017) 57–96, 1st Quart.
- [90] S. Zhang, et al., 1.5 Gbit/s multi-channel visible light communications using CMOS-controlled GaN-based LEDs, *J. Lightw. Technol.* 31 (8) (2013) 1211–1216. Apr. 15.
- [91] A.S. Ghazy, H.A.I. Selmy, H.M.H. Shalaby, Fair resource allocation schemes for cooperative dynamic free-space optical networks, *IEEE/OSA J. Opt. Commun. Netw.* 8 (11) (2016) 822–834. Nov.
- [92] M.P. Peloso, I. Gerhardt, Statistical tests of randomness on quantum keys distributed through a free-space channel coupled to daylight noise, *J. Lightw. Technol.* 31 (23) (2013) 3794–3805. Dec. 1.
- [93] W.J. Miniscalco, S.A. Lane, Optical space-time division multiple access, *J. Lightw. Technol.* 30 (11) (2012) 1771–1785. Jun. 1.
- [94] K. Su, L. Moeller, R.B. Barat, J.F. Federici, Experimental comparison of performance degradation from terahertz and infrared wireless links in fog, *J. Opt. Soc. Amer. A, Opt. Image Sci.* 29 (2) (2012) 179–184. Feb.
- [95] H.A. Willebrand, B.S. Ghuman, Fiber optics without fiber, *IEEE Spectr* 40 (8) (2001) 41–45. Aug.
- [96] E. Leitgeb, et al., Current optical technologies for wireless access, in: *Proc. Int. Conf. Telecommun.*, 2009, pp. 7–17. Jun.
- [97] D. Kedar, S. Arnon, Urban optical wireless communication networks: the main challenges and possible solutions, *IEEE Commun. Mag.* 42 (5) (2004) S2–S7. May.
- [98] Huy Nguyen, Byung Deok Chung, Yeong Min Jang, Deep Learning based RS-OFDM scheme considering mobility environment for Optical Camera Communication system, in: *2023 Fourteenth International Conference on Ubiquitous and Future Networks (ICUFN)*, . 2023. July.
- [99] Huy Nguyen, Duc Hoang Tran, ByungDeok Chung Van Linh Nguyen, Yeong Min Jang, Design and Implementation of RS-OFDM scheme for Optical Camera Communication based on Deep Learning, in: *2023 International Conference on Artificial Intelligence in Information and Communication (ICAICI)*, 2023. Feb.
- [100] Huy Nguyen, Yeong Min Jang, Design and Implementation of Rolling Shutter MIMO-OFDM scheme for Optical Camera Communication System, in: *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, 2021. Oct.
- [101] C. Liu, X.-G. Xia, Y. Li, X. Gao, H. Zhang, Omnidirectional Quasi-Orthogonal Space-Time Block Coded Massive MIMO Systems, *IEEE Communications Letters* 23 (9) (2019) 1621–1625. Sept.
- [102] S.R. Teli, V.M. Icaza, S. Vitek, R. Perez-Jimenez, Z. Ghassemloooy, S. Zvanovec, FFT-Based Frequency Detection and Interference Analysis for NLOS MIMO-OCC, *IEEE Access* 12 (2024) 99137–99148.
- [103] M. Abdrabou, T.A. Gulliver, Adaptive Physical-Layer Authentication for IoT in MIMO Communication Systems Using Support Vector Machine, *IEEE Internet of Things Journal* 10 (22) (2023) 19861–19873, 15 Nov.15.
- [104] M. Elbayoumi, M. Kamel, W. Hamouda, A. Youssef, NOMA-Assisted Machine-Type Communications in UDN: state-of-the-Art and Challenges, *IEEE Communications Surveys & Tutorials* 22 (2) (2020) 1276–1304. Secondquarter.
- [105] S.A.H. Mohsan, et al., NOMA-Based VLC Systems: a Comprehensive Review, *Sensors* 23 (8) (2023) 7815.
- [106] Y. Noma, W. Chujo, Line-of-sight distance extension of rolling-shutter OCC using dual-camera spatial luminance distribution. 2020 IEEE Globecom Workshops, GC Wkshps, Taipei, Taiwan, 2020, pp. 1–6.
- [107] Tung Lam Pham, Huy Nguyen, Trang Nguyen, Yeong Min Jang “A Novel Neural Network-Based Method for Decoding and Detecting of the DS8-PSK Scheme in an OCC System, *MDPI Applied Sciences* (2019).

- [108] Z.-R. Zhu, J. Zhang, R.-H. Chen, H.-Y. Yu, Autoencoder-Based Transceiver Design for OWC Systems in Log-Normal Fading Channel, *IEEE Photonics J* 11 (5) (. 2019) 1–12. Oct.
- [109] Huy Nguyen, Yeong Min Jang, Design and Implementation of Deep Learning-Based MIMO C-OOK Scheme for Optical Camera Communication Considering Mobility Environment, in: 2024 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC), 2024. Feb.
- [110] M.A. Arfaoui, et al., Invoking Deep Learning for Joint Estimation of Indoor LiFi User Position and Orientation, *IEEE Journal on Selected Areas in Communications* 39 (9) (. 2021) 2890–2905. Sept.
- [111] With Ranges up to 200 m, for Both Stationary and Mobile Devices, in: *IEEE Standard For Multi-Gigabit per Second Optical Wireless Communications (OWC)*, 802, IEEE Std, 2023, p. 13. -2023.
- [112] *IEEE Standard for Information Technology–Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks-Specific Requirements Part 11: wireless Lan Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 6: light Communications*, IEEE Standard 802.11bb-2023 (Amendment to IEEE Standard 802.11-2020 as amended by IEEE Standard 802.11ax-2021, IEEE Standard (2023), 802.11ay-2021, IEEE Standard 802.11ba-2021, IEEE Standard 802.11az-2022, IEEE Standard 802.11-2020/Cor 1- 2022, and IEEE Standard 802.11bd-2023).
- [113] Rajesh Uppal, *Illuminating the Future: The Rising Role of Laser Free Space Optical Communication (FSO) (2024)*. Accessed: Nov. 2024. [Online]. Available, <https://idstch.com/space/illuminating-the-future-the-rising-role-of-laser-free-space-optical-communication-fso/>.
- [114] X. Fang, et al., Joint Communication and Sensing Toward 6G: models and Potential of Using MIMO, *IEEE Internet Things J* 10 (. 2023) 4093–4116. Mar.
- [115] Y. Ata, H. Abumarshoud, et al., Intelligent reflecting surfaces for underwater visible light communications, *IEEE Photon. J.* 15 (1) (. 2023) 1–10. Feb.
- [116] A. Gupta, et al., FD-OMAC: a Novel Full-Duplex Optical Media Access Control for IEEE 802.15.7, *IEEE Access* 9 (2021) 148894–148910. Nov.
- [117] M.M. Azari, et al., Evolution of Non-Terrestrial Networks From 5G to 6G: a Survey, *IEEE Communications Surveys & Tutorials* 24 (4) (2022) 2633–2672. Fourthquarter.